

Software Requirements Specification

Digital Knowledge Platform

[DU Ironclad]

Version: 1.0
Status: Draft
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A comprehensive platform integrating Digital Archive, Research Repository,
Library Management, and Student Project Showcase

April 9, 2026

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Chapter 1

Document Control

1.1 Metadata

- **Team:** DU_Ironclad
- **Project Name:** Digital Knowledge Platform
- **Repository URL:** [To be added]
- **Version:** 1.0
- **Status:** Draft
- **Date:** March 7, 2026

1.2 Authors and Reviewers

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Chapter 2

Problem Definition

2.1 Problem Statement

2.1.1 What problem are we solving?

Academic institutions face significant challenges in managing and providing access to their digital knowledge resources. Current systems are fragmented, with separate platforms for library management, research repositories, digital archives, and student project showcases. This fragmentation leads to:

- Difficulty in discovering relevant academic resources
- Lack of proper research output visibility and impact tracking
- Inefficient library operations and poor user experience
- Limited support for Bangla language content and bilingual research
- No collaborative tools for knowledge sharing and annotation
- Poor preservation of institutional knowledge and historical documents
- Absence of advanced analytics on research trends and usage patterns
- Limited academic integrity checking mechanisms

2.1.2 Why does it matter?

A unified Digital Knowledge Platform is essential for:

- Enhancing research productivity and collaboration
- Improving accessibility of knowledge resources for students and faculty
- Preserving institutional heritage and research outputs
- Supporting bilingual (Bangla-English) academic work
- Providing data-driven insights for academic planning
- Streamlining library operations and reducing administrative overhead
- Fostering academic integrity and quality research outputs

2.2 Target Users

User Group	Description	Main Needs
Students	Undergraduate and graduate students	Access to library resources, research papers, project showcases, collaborative annotation, citation management
Researchers/Faculty	Academic staff conducting research	Publishing research outputs, advanced search, AI-powered research assistance, citation network analysis, collaborative tools
Librarians	Library staff managing collections	Efficient CRUD operations, bulk import/export, lending management, overdue tracking, reporting, barcode/QR scanning
Archive Managers	Staff managing institutional archives	Document upload with metadata, versioning, access control, digital preservation, format migration
Administrators	System administrators and institutional leadership	Analytics dashboard, user management, access tier configuration, system monitoring
General Public	External visitors	Public search and browse, viewing published content, research discovery
Lab/Department Heads	Academic unit leaders	Lab portfolio management, student project showcase, contribution tracking, trend analysis

2.3 Goals and Non-Goals

2.3.1 Goals

- Create a unified platform integrating digital archive, research repository, library catalog, and project showcase
- Support multi-format media (PDF, DOCX, audio, video, images) with rich metadata
- Implement bilingual (Bangla-English) support with advanced NLP features
- Provide AI-powered research assistance and smart recommendations
- Enable collaborative annotation and knowledge sharing
- Offer comprehensive library management with lending/return workflow
- Implement tiered access control (public/member/staff/restricted)
- Provide analytics dashboard for research trends and usage patterns

- Ensure academic integrity through plagiarism detection and citation checking
- Support digital preservation with format health monitoring
- Create virtual reading rooms for collaborative study
- Implement smart citation management with multiple format support

2.3.2 Non-goals

- Course management or LMS features (covered by existing systems)
- Financial transactions or e-commerce (except fine management)
- Video conferencing infrastructure (integrate with existing tools)
- Email server or communication platform
- Mobile app development (initial version is web-based with responsive design)
- Content creation tools (editors, graphic design software)

Chapter 3

Scope and Context

3.1 In Scope

3.1.1 Core Modules

Digital Archive System

- Document/media upload with rich metadata editor
- Multi-format support (PDF, DOCX, PPTX, audio, video, images)
- Versioning and lifecycle management (draft/review/published/archived)
- Bulk upload and metadata editing tools
- Digital preservation system with integrity monitoring
- Bangla OCR for scanned documents

Research Repository

- Research output submission and publication
- Lab portfolio and department research showcase
- AI-powered research assistant (summarization, key findings extraction)
- Citation network visualization
- Academic integrity suite (plagiarism detection, citation checker)
- Smart citation manager with multiple format support

Student Project Showcase

- Project submission with multimedia support
- Project categorization and tagging
- Peer review and feedback system
- Achievement badges and recognition

Digital Library & Catalog

- Advanced catalog search (Bangla and English)
- Lending/return workflow with due tracking
- Member account features (history, wishlist, hold requests)
- Barcode/QR support for physical item operations
- Overdue and fine management
- Librarian dashboard with CRUD, import, and reporting

Collaborative Features

- Real-time collaborative PDF/document annotation
- Discussion threads on document sections
- Virtual reading rooms with study groups
- Shared annotated versions

Analytics & Intelligence

- Interactive analytics dashboard
- Research trend visualization
- Citation impact metrics
- Reading patterns and popular topics
- Department contribution leaderboards

Advanced Language Support

- Bangla-English bilingual interface
- Bangla NLP with stemming and synonym support
- Translation integration
- Voice search in both languages

3.2 Out of Scope

- Course content delivery and LMS features
- Student grade management
- Video conferencing platform development
- Native mobile applications (Phase 1)

- Payment gateway integration for fines (cash/manual tracking only)
- Hardware procurement for barcode scanners
- Automated email server setup
- Third-party content acquisition negotiations

3.3 Assumptions and Constraints

3.3.1 Technical Constraints

- Web-based responsive application (desktop and mobile browsers)
- Must support modern browsers (Chrome, Firefox, Safari, Edge - last 2 versions)
- Maximum file upload size: 500MB per file
- Server infrastructure availability (provided by institution)
- MVP core features shall function without external cloud API dependencies
- Optional cloud service integrations (Phase 2) require institutional Memoranda of Understanding (MoU) and network access policies
- If cloud APIs are used, implement graceful fallback to self-hosted alternatives

3.3.2 Organizational Constraints

- Access to institutional data for migration
- Availability of existing library database
- User training resources for librarians and staff
- Institutional policies on data privacy and access control

3.3.3 Timeline Constraints

- SRS completion: Lab 2
- Peer review: Lab 3
- Implementation of Sprint 1,2: Lab 4,5
- Peer UI critique: Lab 6
- Testing baseline: Lab 7
- Peer QA: Lab 8
- Hardening and Deployment Readiness: Lab 9
- Testing and deployment by semester end

3.3.4 Assumptions

- Users have basic computer literacy
- Internet connectivity available
- Institution provides server resources and backup systems
- Existing library data can be exported in standard formats (CSV, MARC)
- Copyright clearance obtained for uploaded content
- Users have valid institutional email addresses

Chapter 4

Stakeholders and Roles

4.1 Stakeholders

Stakeholder	Interest	Priority
Students and Researchers	Access to knowledge resources, research tools, collaboration features	High
Library Staff	Efficient operations, better user service, reduced manual work	High
Faculty Members	Research visibility, citation tracking, teaching resources	High
IT Department	System integration, security, maintenance	High
University Administration	Cost efficiency, academic reputation, compliance	Medium
Archive Department	Preservation of institutional knowledge, accessibility	Medium
External Researchers	Access to public research outputs	Low
General Public	Access to open knowledge resources	Low

4.2 User Roles and Permissions

Role	Description	Permissions
Guest	Unauthenticated visitor	Browse public content, basic search
Member	Registered student/faculty	All guest permissions + borrow items, access member content, create annotations, join reading rooms, save to wishlist
Contributor	Author/researcher	All member permissions + submit research/projects, upload to repository, draft content
Staff	Librarian/archive staff	All member permissions + content review, metadata editing, lending management, bulk operations, access all content tiers
Lab Manager	Department/lab head	All contributor permissions + manage lab portfolio, approve lab submissions, view lab analytics

Role	Description	Permissions
Administrator	System admin	Full system access, user management, system configuration, analytics, access tier management
Reviewer	Peer reviewer	Review assigned submissions, provide feedback, recommend publication

Chapter 5

Functional Requirements

Requirement ID format: FR-DKP-###

Priority Legend:

- **M (Must):** Critical for basic operation
- **S (Should):** Important for competitive advantage
- **C (Could):** Nice to have, enhances user experience
- **W (Won't):** Not in this release

5.1 Authentication and Access Control

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-001	User Authentication	System shall provide secure user authentication with institutional email	M	Users can register, login, reset password; sessions expire after 30 min inactivity
FR-DKP-002	Role-Based Access	System shall enforce role-based permissions for all operations	M	Users only access features/content per their assigned role

5.2 Document Management

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-003	Document Upload	System shall support upload of PDF, DOCX, PPTX, MP4, MP3, JPG, PNG files up to 500MB	M	Files upload successfully with progress indicator; unsupported formats rejected

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-004	Metadata Editor	System shall provide rich metadata editor with at least 20 fields	M	All metadata fields editable; validation for required fields; auto-save every 30 seconds
FR-DKP-005	Bulk Upload	System shall support bulk upload via CSV/Excel with metadata	S	Upload up to 100 files at once; validation report generated
FR-DKP-006	Versioning	System shall maintain version history for all documents	M	Each edit creates new version; view/restore previous versions
FR-DKP-007	Lifecycle States	System shall support draft, review, published, archived states	M	Only authorized users can change states
FR-DKP-008	Access Tiers	System shall implement public, member, staff, restricted access tiers	M	Content shows only to authorized users

5.3 Search and Language Support

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-009	Bilingual Search	System shall provide search in both Bangla and English with filters	M	Search returns relevant results; filters by date, type, author, category
FR-DKP-010	Advanced Bangla NLP	System shall support Bangla text with stemming, synonym search	S	Bangla search handles variations; suggests corrections

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-011	Bangla OCR	System shall extract text from scanned Bangla documents with measurable accuracy	S	OCR processes Bengali PDFs/images; Character Error Rate (CER) $\leq 15\%$ on BanglaOCR benchmark dataset; manual correction UI provided
FR-DKP-012	Voice Search	System shall support voice search in Bangla and English	C	Voice input converts to text; search executes
FR-DKP-013	Translation	System shall provide Bangla-English translation for abstracts	C	Translate button available; translation displays in overlay
FR-DKP-014	Media Viewer	System shall provide in-browser viewer for all supported formats	M	PDF, DOCX render in browser; video/audio player with controls

5.4 Library Management

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-015	Catalog Search	System shall provide advanced library catalog search with faceted filters	M	Search by title, author, ISBN, subject; filters by availability, location
FR-DKP-016	Lending Workflow	System shall support item checkout, renewal, and return with due dates	M	Members can borrow available items; renewal up to 2 times
FR-DKP-017	Due Tracking	System shall track borrowed items and send overdue notifications	M	Email notifications at 3 days, 1 day before, and daily after due date

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-018	Fine Management	System shall calculate and track overdue fines	S	Fines calculated per policy; displayed in member account
FR-DKP-019	Hold Requests	Members shall be able to place hold requests on checked-out items	S	Hold request queued; notification when item available
FR-DKP-020	Wishlist	Members shall be able to create and manage wishlist	C	Add/remove items from wishlist; email on status changes
FR-DKP-021	Borrowing History	Members shall view their complete borrowing history	S	Display all past and current borrows; export to CSV/PDF
FR-DKP-022	Barcode/QR Scan	System shall support barcode and QR code scanning for item operations	S	Scan barcode/QR to identify item; works with standard scanners
FR-DKP-023	Librarian CRUD	Librarians shall have full CRUD access to catalog items	M	Create, read, update, delete catalog entries; audit log maintained
FR-DKP-024	Bulk Import	System shall support bulk import of catalog data from MARC, CSV	M	Import up to 10,000 records; validation report; duplicate detection
FR-DKP-025	Librarian Reports	System shall generate reports on circulation, popular items, overdue	M	Generate 10+ report types; export to PDF/Excel; date range filters

5.5 Research Repository

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-026	Research Submission	Researchers shall submit papers, theses, datasets with metadata	M	Multi-step form; plagiarism check; assigns identifier

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-027	Lab Portfolio	Lab managers shall create and manage lab research portfolio	S	Add lab info, members, projects, publications; analytics
FR-DKP-028	Project Showcase	Students shall submit projects with multimedia for showcase	M	Upload files, images, videos; description up to 5000 chars
FR-DKP-029	AI Summarization	System shall auto-generate summaries of research papers with ROUGE-L evaluation	S	Extract abstract; generate 200-word summary; ROUGE-L score ≥ 0.35 on CNN/DailyMail or academic paper benchmark; user can accept/reject/edit
FR-DKP-030	Key Findings Extraction	System shall extract key findings and contributions from papers	C	Identify methodology, results, conclusions; editable by author
FR-DKP-031	Smart Recommendations	System shall recommend related content based on reading patterns	S	Analyze user activity; recommend 5-10 similar items
FR-DKP-032	Citation Network	System shall visualize citation relationships between papers	S	Interactive graph of citations; filter by year, author, lab
FR-DKP-033	Plagiarism Detection	System shall check uploaded documents for plagiarism	M	Compare against repository; similarity report with highlights
FR-DKP-034	Citation Checker	System shall verify citation completeness and formatting	S	Scan document for citations; check consistency; suggest fixes
FR-DKP-035	Smart Citation Manager	System shall generate citations in multiple formats	M	Auto-extract data; format in APA, MLA, IEEE, Chicago, Vancouver

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-036	Reference Export	System shall export citations to Zotero, Mendeley, End-Note formats	C	Support RIS, BibTeX, End-Note XML; batch export up to 100 items

5.6 Collaborative Features

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-037	Collaborative Annotation	Users shall annotate documents and share with others	S	Highlight text, add comments; threading; permissions control
FR-DKP-038	Discussion Threads	Users shall create discussions on document sections	S	Click-to-discuss on paragraphs; threaded replies; notifications
FR-DKP-039	Annotation Export	Users shall export annotations separately	C	Export annotations to PDF, Word, text; include context
FR-DKP-040	Virtual Reading Rooms	Users shall create study groups for collaborative reading	C	Create room; invite members; real-time presence; chat

5.7 Analytics and Digital Preservation

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-041	Analytics Dashboard	Admins shall view interactive analytics on platform usage	M	15+ metrics: visits, downloads, popular items, trends
FR-DKP-042	Research Trends	System shall visualize research trends over time	S	Topic analysis; trend graphs; emerging areas
FR-DKP-043	Citation Metrics	System shall calculate and display citation impact metrics	S	Citation count, h-index, impact factor; author rankings

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-DKP-044	Reading Patterns	System shall analyze and display reading patterns	C	Time-of-day usage; popular categories; session duration
FR-DKP-045	Contribution Leaderboards	System shall display top contributors by department/lab	C	Rank by publications, citations, views; gamification
FR-DKP-046	Digital Preservation	System shall monitor file format health and integrity	S	Checksum verification daily; format obsolescence warnings
FR-DKP-047	Format Migration	System shall suggest and support format migrations	C	Identify at-risk formats; semi-automated conversion
FR-DKP-048	Automated Backup	System shall perform automated backups with snapshots	M	Daily incremental, weekly full backups; 30-day retention
FR-DKP-049	Notification System	System shall send email notifications for key events	M	Configurable alerts: due dates, holds, new content
FR-DKP-050	Activity Log	System shall maintain audit logs of all significant actions	M	Log user actions with timestamp; 1-year retention

Chapter 6

User Stories and Use Cases

6.1 User Stories

Minimum 12 user stories required - 20 provided

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-DKP-001	Student	browse the digital archive with bilingual search	I can find relevant academic resources in my preferred language	Search interface in Bangla/English; results displayed with thumbnails
US-DKP-002	Researcher	submit my research paper to the repository	my work is preserved and discoverable by others	Multi-step submission form; metadata auto-filled; confirmation email sent
US-DKP-003	Librarian	scan a book's barcode to check it out to a member	I can process loans quickly at the desk	Barcode scan identifies item; checkout completes in < 10 seconds
US-DKP-004	Student	annotate PDF documents and share with my study group	we can collaborate on understanding complex papers	Highlight and comment tools; share link to annotated version
US-DKP-005	Faculty	see citation network visualization for my publications	I can understand my research impact and connections	Interactive graph loads; shows citing and cited papers
US-DKP-006	Archive Manager	upload historical documents in bulk with metadata	I can efficiently digitize our institutional heritage	CSV template provided; 100+ files upload together
US-DKP-007	Student	receive AI-generated summaries of research papers	I can quickly assess paper relevance before full read	Summary appears on paper detail page; accuracy > 80%
US-DKP-008	Librarian	generate a report of overdue items and fines	I can send reminders and manage collections	Report filters by date range; export to Excel
US-DKP-009	Researcher	get automatic citation formatting in APA style	I can easily reference works in my writing	Copy citation button; APA format displayed

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-DKP-010	Lab Manager	showcase my lab's research portfolio	prospective students and collaborators can see our work	Lab page shows projects, publications, members
US-DKP-011	Student	place a hold on a checked-out book	I get notified when it becomes available	Hold request button; queue position shown; email when available
US-DKP-012	Administrator	view analytics on research trends across departments	I can make informed decisions on resource allocation	Dashboard shows trending topics; comparative charts; quarterly snapshots
US-DKP-013	Member	search using voice in Bangla	I can find resources hands-free in my native language	Microphone button; speech-to-text in Bangla; results relevant
US-DKP-014	Researcher	check my paper for plagiarism before submission	I can ensure academic integrity and originality	Similarity report in 2-3 minutes; sources highlighted; percentage shown
US-DKP-015	Student	submit my final year project to the showcase	my work can inspire future students and be archived	Submission form; multiple file upload; approval workflow
US-DKP-016	Librarian	use QR codes to identify damaged books	I can quickly update item status without typing	QR scan with mobile; item details shown; status update saves
US-DKP-017	Faculty	export my publication list to End-Note	I can manage my references in my preferred tool	Export button; End-Note XML format; all metadata included
US-DKP-018	Member	create a virtual reading room for exam prep	my study group can collaborate on course materials	Create room; invite via link; shared viewing; chat available
US-DKP-019	Archive Staff	monitor digital preservation health of archived files	I can proactively address format obsolescence	Dashboard shows format warnings; checksum status; migration suggestions
US-DKP-020	Student	see recommended papers based on what I'm reading	I can discover related research efficiently	Recommendations in sidebar; machine learning based; refresh periodically

6.2 Critical Use Cases

6.2.1 Use Case 1: Research Paper Submission with AI Assistance

Actor: Researcher (Faculty/Graduate Student)

Preconditions:

- User is logged in with Contributor or higher role
- User has a research paper file ready (PDF or DOCX)
- Paper meets minimum quality standards

Main Flow:

1. User navigates to "Research Repository"

>

"Submit Paper"

2. System displays multi-step submission wizard
3. User uploads paper file (validation: format, size, virus scan)
4. System extracts metadata automatically (title, authors, abstract, keywords)
5. User reviews and edits metadata in rich editor
6. System runs plagiarism detection in background
7. User selects lab/department association and research area
8. System generates AI summary and key findings
9. User reviews and edits AI-generated content
10. User selects access tier (public/member/staff)
11. System assigns unique identifier (DKP-YEAR-XXXX)
12. User submits for review with optional reviewers
13. System sends confirmation email and notifies reviewers
14. Paper enters "Under Review" state

Alternate Flows:

- **3a.** File validation fails → Show error message; allow retry
- **6a.** Plagiarism score

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25% → Display warning; highlight similar sections

- **8a.** AI summary fails → Allow manual summary; log error

- **12a.** User saves as draft → Paper saved in "Draft" state

Postconditions:

- Paper is stored in repository
- Metadata is searchable
- Audit log entry created
- Notifications sent

6.2.2 Use Case 2: Book Checkout with Barcode/QRcode Scanning

Actor: Librarian

Preconditions:

- Librarian is logged in at circulation desk
- Member is present with ID
- Item is available for checkout
- Barcode/QRcode scanner is connected

Main Flow:

1. Librarian scans member ID barcode/QR code
2. System displays member profile with photo and account status
3. System checks for holds, overdue, and outstanding fines
4. Librarian scans book barcode
5. System retrieves book details and availability status
6. System calculates due date based on item type and member category
7. System creates checkout record
8. System updates item status to "Checked Out"
9. System displays checkout confirmation with due date
10. System sends confirmation email to member
11. Librarian hands book and receipt to member

Alternate Flows:

- **3a.** Member has overdue items → Display overdue list; allow override if fine threshold
- **3b.** Member has outstanding fines 500tk → Block checkout; require payment
- **5a.** Book already checked out → Suggest placing hold

Postconditions:

- Checkout record created with timestamp
- Item status updated to "Checked Out"
- Due date scheduled in system calendar
- Member's account updated
- Notification email sent

6.2.3 Use Case 3: Collaborative Document Annotation

Actor: Student (Member role)

Preconditions:

- User is logged in
- User has access to the document
- Document is in a supported format (PDF)
- User is member of at least one study group

Main Flow:

1. User opens research paper in document viewer
2. System loads PDF with annotation toolbar visible
3. User selects text passage to annotate
4. User clicks "Add Note" from toolbar
5. System displays annotation editor overlay
6. User types comment/observation (rich text supported)
7. User tags annotation with category (question/insight/critique)
8. User selects visibility: Private / Study Group / Public
9. User saves annotation
10. System stores annotation linked to document location
11. System displays annotation as highlighted text with icon
12. User clicks "Share Annotated Version"
13. System generates unique link for study group
14. System sends notification to group members
15. Group members access same document with shared annotations

16. Members can reply to annotations creating discussion thread

Alternate Flows:

- **3a.** User wants to highlight without comment → Select text; choose color; save without text
- **8a.** User wants to tag specific members → Type @ and select members; they receive notification
- **12a.** User wants to export annotations → Select "Export Annotations"; choose format; download

Postconditions:

- Annotations saved to database
- Document view updated in real-time for all viewers
- Activity feed updated for group
- Search index updated (annotations are searchable)

6.2.4 Use Case 4: Digital Preservation Health Check

Actor: Archive Manager (Staff role)

Preconditions:

- User has Staff or Admin role
- Digital preservation system is configured
- Archive contains documents of various ages and formats

Main Flow:

1. Archive Manager navigates to "Digital Preservation Dashboard"
2. System displays health overview: total files, checksums verified, at-risk formats
3. User filters by collection or date range
4. System runs automated health check on selected files
5. System verifies checksums for file integrity
6. System checks file formats against obsolescence database
7. System generates health report with color-coded status
8. User reviews files flagged as "At Risk" (e.g., WordPerfect, Flash)
9. User selects flagged files for migration
10. System recommends modern format equivalents

11. User approves format migration plan
12. System creates migration queue and schedules conversion jobs
13. System converts files while preserving originals
14. System validates converted files for completeness
15. User reviews sample conversions for quality
16. User approves batch migration
17. System updates metadata and links to new versions
18. System archives original formats in preservation tier
19. System sends completion report to Archive Manager

Alternate Flows:

- **5a.** Checksum mismatch detected → Alert user; flag file as corrupted; restore from backup
- **10a.** No clear format equivalent → User specifies custom converter; manual review required
- **14a.** Conversion fails for some files → Log failures; retry with alternative converter

Postconditions:

- File integrity verified for all checked files
- At-risk files identified and migration planned/completed
- Metadata updated with preservation actions
- Audit log records all preservation activities

6.2.5 Use Case 5: AI-Powered Research Discovery

Actor: Faculty Researcher

Preconditions:

- User is logged in
- User has previously viewed/downloaded papers (history exists)
- AI recommendation engine is trained
- Repository contains sufficient papers for analysis

Main Flow:

1. User logs into platform
2. System analyzes user's reading history and research profile

3. System identifies user's research interests from metadata
4. User navigates to "Recommended for You" section
5. System displays 10 recommended papers with relevance scores
6. User clicks on a recommended paper
7. System displays paper details with AI-generated summary
8. System highlights key findings relevant to user's interests
9. System shows citation network visualization
10. User explores cited and citing papers
11. User clicks on related paper from network
12. System extracts common themes and displays them
13. User saves multiple papers to reading list
14. System updates recommendation model with new interaction
15. User exports citation list in APA format

Alternate Flows:

- **5a.** No recommendations available (new user) → Show trending papers; ask user to select interests
- **7a.** AI summary not available → Display abstract; offer to generate summary (1-2 min)
- **9a.** Paper has no citations yet → Show similar papers by topic

Postconditions:

- User's reading list updated
- Recommendation model retrained with new interaction data
- User activity logged for analytics
- Citation export completed

Chapter 7

Data Requirements

7.1 Core Entities

Entity	Description	Key Fields	Constraints
User	People using the platform	user_id (PK), email, name, role, department, status	Email unique; role from enum
Document	Digital content items	doc_id (PK), title, type, format, file_path, version, state	File path must exist; state from lifecycle enum
Metadata	Rich metadata for documents	metadata_id (PK), doc_id (FK), author, keywords, language, abstract	JSON structured fields
CatalogItem	Library items (physical and digital)	item_id (PK), title, isbn, barcode, location, status	Barcode unique; status from enum
Loan	Borrowing transactions	loan_id (PK), item_id (FK), member_id (FK), checkout_date, due_date	Due date & check-out date
Annotation	User annotations on documents	annotation_id (PK), doc_id (FK), user_id (FK), location, text, visibility	Location as page+coordinates
ResearchPaper	Scholarly publications	paper_id (PK), doc_id (FK), authors, citations_count, lab_id (FK)	Authors as JSON array
Lab	Research labs/departments	lab_id (PK), name, head_id (FK), description	Head must be faculty role

(Full document contains 14 core entities including Project, Citation, ReadingRoom, Analytics, Format, and Preservation)

7.2 Data Dictionary

Representative sample of key fields:

Field	Type	Required	Validation/Constraint	Example
user_id	UUID	Yes	Auto-generated, PK	"550e8400-e29b..."
email	VARCHAR(255)	Yes	Valid email, unique, institutional domain	"john.doe@university.edu"
role	ENUM	Yes	guest, member, contributor, staff, admin, reviewer	"member"
password_hash	VARCHAR(255)	Yes	Bcrypt hash, min 8 chars	"\$2b\$12\$KIX..."
title	VARCHAR(500)	Yes	Min 3 chars, max 500	"Machine Learning..."
file_size	BIGINT	Yes	Bytes, max 524,288,000 (500MB)	2048576
state	ENUM	Yes	draft, review, published, archived	"published"
access_tier	ENUM	Yes	public, member, staff, restricted	"member"
barcode	VARCHAR(50)	No	Unique, alphanumeric	"BC123456789"
due_date	TIMESTAMP	Conditional	Required for active loans, future date	"2026-03-25 23:59:59"

7.3 Data Model

Entity Relationship Overview:

```

USER (1) (M) DOCUMENT [uploads]
USER (1) (M) ANNOTATION [creates]
USER (1) (M) LOAN [borrows]
USER (1) (M) READINGROOM [creates/joins]

DOCUMENT (1) (1) METADATA [describes]
DOCUMENT (1) (M) ANNOTATION [has]
DOCUMENT (1) (M) PRESERVATION [tracked by]
DOCUMENT (1) (1) RESEARCHPAPER [optionally is]

CATALOGITEM (1) (M) LOAN [has]
RESEARCHPAPER (M) (M) CITATION [cites/cited by]
RESEARCHPAPER (M) (1) LAB [belongs to]

```

Key Relationships:

- One-to-many: User uploads many Documents
- One-to-one: Document has exactly one Metadata record
- Many-to-many self-referencing: ResearchPapers cite each other (Citation network)
- Self-referencing: Annotations can have threaded replies (parent_id)

7.3.1 ER Diagram Deliverable Requirement

The text-based model above must be accompanied by ER-DKP-001 as a formal diagram artifact. The diagram shall include complete cardinalities, PK/FK notation, optionality, and relationship names for all core entities.

Chapter 8

UI and UX Requirements

8.1 Wireframes

Note: Detailed wireframes to be created using Figma/Balsamiq. Key screens described below:

8.1.1 Wireframe 1: Home Page / Dashboard

Purpose: Landing page for authenticated users

Key Components:

- Top navigation: Logo, Search bar (with language toggle), Main menu, User profile
- Hero section: Featured content carousel, Quick stats
- Main content: Recommended for You (AI-powered), Trending Research, Recent Activity
- Quick Actions: Upload Document, Submit Paper, Borrow Item
- Footer: Help, About, Contact, Language selector

8.1.2 Wireframe 2: Advanced Search Interface

Purpose: Comprehensive search and filter for finding content

Key Components:

- Left sidebar: Filters (Content Type, Language, Date range, Author, Access tier)
- Main area: Search bar with voice input, Sort options, Results with pagination
- Result cards: Thumbnail, Title, Authors, Abstract snippet, Action buttons
- Right sidebar: Search tips, Recent searches, Saved searches

8.1.3 Wireframe 3: Document Viewer with Annotation Tools

Key Components:

- Top toolbar: Navigation controls, Annotation tools, Share, Download, Cite
- Left panel: Table of contents, Thumbnail page navigator
- Center: PDF render area with existing annotations
- Right panel: Annotations tab, Discussion threads, Document details

(Full document includes 8 wireframes: Research Submission Wizard, Library Catalog Interface, Librarian Dashboard, Analytics Dashboard, and Virtual Reading Room)

8.1.4 Visual Artifact Deliverables

To make this SRS implementation-ready, the following visual artifacts are mandatory deliverables (not optional prose descriptions):

Artifact ID	Artifact	Minimum Content	Format
WF-DKP-001	Home/Dashboard Wire-frame	Navigation, global search, recommendation section, quick actions, responsive behavior notes	Figma/Balsamiq export (PNG/PDF)
WF-DKP-002	Advanced Search Wire-frame	Filters, result ranking, pagination, bilingual input, saved search UX	Figma/Balsamiq export (PNG/PDF)
WF-DKP-003	Document Viewer Wire-frame	Annotation tools, side panels, discussion threads, sharing controls	Figma/Balsamiq export (PNG/PDF)
WF-DKP-004	Research Submission Wizard	Multi-step flow, validation states, plagiarism/summarization progress states	Figma/Balsamiq export (PNG/PDF)
WF-DKP-005	Library Circulation UI	Barcode/QR workflow, checkout/return states, exception handling UI	Figma/Balsamiq export (PNG/PDF)
ER-DKP-001	Logical ER Diagram	Entities, PK/FK, cardinalities, many-to-many bridge tables, lifecycle/status enums	Mermaid/Draw.io/PDF
ARCH-DKP-001	On-Prem Deployment Diagram	Network zones, reverse proxy, app services, DB/search/cache/queue/storage, backup nodes	Draw.io/PDF

8.2 UX and Accessibility Baseline

8.2.1 Core UX Principles

1. **Clarity:** Clear labels, intuitive navigation, consistent terminology
2. **Efficiency:** Minimal clicks, keyboard shortcuts, bulk operations
3. **Feedback:** Loading indicators, success/error messages, progress bars
4. **Forgiveness:** Confirmation dialogs, undo, auto-save drafts
5. **Consistency:** Unified design system, predictable layouts

8.2.2 Accessibility Requirements (WCAG 2.1 Level AA)

- **Keyboard Navigation:** All interactive elements accessible via Tab; skip navigation links; focus indicators (3:1 contrast)
- **Screen Reader Support:** Semantic HTML5; ARIA labels; alt text for images; proper form labels
- **Visual Design:** 4.5:1 contrast ratio for normal text; 3:1 for large text; text resizable to 200%; no color-only information
- **Responsive Design:** Mobile-first approach; breakpoints for Mobile (<768px), Tablet (768-1024px), Desktop (>1024px); 44x44px touch targets
- **Language Support:** Language selector on all pages; Bangla Unicode font support; full UI translation
- **Error Handling:** Inline validation; error summaries; suggestions for fixes
- **Multimedia:** Captions for videos; transcripts for audio; keyboard-accessible controls

Chapter 9

Non-Functional Requirements

NFR ID format: NFR-DKP-###

9.1 Performance Requirements

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-001	Performance	UI page render time for standard pages (excluding long-running AI/OCR jobs)	≤ 2 seconds on 10 Mbps
NFR-DKP-002	Performance	Search results response time	≤ 1 second for up to 1000 results
NFR-DKP-003	Performance	Document viewer initial render time	≤ 3 seconds for PDFs up to 50MB
NFR-DKP-004	Performance	Concurrent user support and API responsiveness	500 concurrent users without degradation; API 95th percentile ≤ 500ms (read), ≤ 2s (write)
NFR-DKP-005	Performance	AI/OCR processing model	Long-running OCR, summarization, and plagiarism checks shall run asynchronously via background queue
NFR-DKP-006	Performance	AI/OCR queue SLA and user feedback	OCR ≤ 2 minutes, summarization ≤ 30 seconds, plagiarism check ≤ 3 minutes; show progress indicator and notify user on completion
NFR-DKP-007	Performance	Database query optimization	No query exceeds 100ms; proper indexing on all FKs

9.2 Scalability Requirements

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-008	Scalability	Horizontal scaling	Architecture supports adding application servers without code changes
NFR-DKP-009	Scalability	Database growth	Support up to 1 million documents and 10 million annotations
NFR-DKP-010	Scalability	Storage capacity	Plan for 10TB initial storage with expansion capability

9.3 Security Requirements

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-011	Security	Authentication	MFA support; password min 8 chars with complexity
NFR-DKP-012	Security	Authorization	RBAC enforced at API and DB levels
NFR-DKP-013	Security	Data encryption	TLS 1.3 in transit; AES-256 at rest
NFR-DKP-014	Security	Session management	Sessions expire after 30 min inactivity; secure tokens
NFR-DKP-015	Security	Input validation	All inputs validated; prevent SQL injection, XSS
NFR-DKP-016	Security	Audit logging	All sensitive operations logged with user, timestamp, action, IP
NFR-DKP-017	Security	File upload security	Virus scanning; MIME verification; size limits
NFR-DKP-018	Security	API security	Rate limiting (100 requests/min per user); API key authentication

9.4 Reliability and Usability

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-019	Reliability	Uptime	99.5% during business hours (7AM-11PM)
NFR-DKP-021	Reliability	Data backup	Daily automated backups; monthly restore tests
NFR-DKP-024	Usability	Learning curve	Basic tasks completable in 5 min without training
NFR-DKP-025	Usability	User satisfaction	≥ 80% satisfaction score in surveys
NFR-DKP-020	Reliability	Mean time to recovery	≤ 4 hours for critical system failures
NFR-DKP-022	Reliability	Data integrity	Checksums verified daily; corruption detection and alerting
NFR-DKP-023	Reliability	Error handling	Graceful degradation; no data loss on errors; user-friendly messages
NFR-DKP-026	Usability	Help and documentation	Context-sensitive help on all pages; comprehensive user guide
NFR-DKP-027	Usability	Error recovery	Users can undo recent actions; clear guidance on error recovery
NFR-DKP-028	Accessibility	WCAG compliance	WCAG 2.1 Level AA verified
NFR-DKP-029	Accessibility	Keyboard navigation	All features accessible via keyboard; logical tab order
NFR-DKP-030	Accessibility	Screen reader support	Compatible with NVDA, JAWS, VoiceOver screen readers
NFR-DKP-031	Accessibility	Visual impairment support	Text zoom up to 200%; high contrast mode

9.5 Compatibility and Maintainability

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-032	Compatibility	Browser support	Full support for Chrome, Firefox, Safari, Edge (last 2 versions)
NFR-DKP-033	Compatibility	Mobile responsiveness	Fully functional on devices from 320px to 2560px width
NFR-DKP-034	Compatibility	Operating systems	Works on Windows 10+, macOS 10.15+, Ubuntu 20.04+, iOS 13+, Android 9+
NFR-DKP-035	Compatibility	Data formats	Import/export support for CSV, Excel, MARC, BibTeX, RIS, EndNote XML
NFR-DKP-036	Maintainability	Code quality	All code passes linting; minimum 80% test coverage; documented APIs
NFR-DKP-037	Maintainability	Modularity	Microservices or layered architecture; clear separation of concerns
NFR-DKP-038	Maintainability	Documentation	Comprehensive technical documentation; architecture diagrams
NFR-DKP-039	Maintainability	Logging and monitoring	Application logs all errors; performance metrics monitored

9.6 Localization and Compliance

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-040	Localization	Language support	Full UI and content support for Bangla and English
NFR-DKP-041	Localization	Unicode support	Proper rendering of Bangla Unicode characters; font selection
NFR-DKP-042	Localization	Date/time formats	Display dates in local format; timezone support

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-043	Compliance	Data privacy and right-to-deletion	User personal data deletable on GDPR request (profile, documents, settings); audit logs retained 1 year with user identifiers anonymized (hashed) and direct identifiable data removed
NFR-DKP-044	Compliance	Copyright compliance	Watermarking for sensitive documents; DRM support
NFR-DKP-045	Compliance	Academic integrity	Plagiarism detection mandatory; audit trail for all edits

9.7 Portability and Extensibility

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-046	Portability	Cloud-agnostic	Can deploy on AWS, Azure, Google Cloud, or on-premise servers
NFR-DKP-047	Portability	Database flexibility	Abstraction layer allows switching databases (PostgreSQL, MySQL)
NFR-DKP-048	Extensibility	Plugin architecture	Support for third-party plugins (OCR providers, translation services)
NFR-DKP-049	Extensibility	API availability	RESTful API for all major operations; GraphQL for complex queries
NFR-DKP-050	Environmental	Energy efficiency	Optimized queries and caching to minimize server load

NFR ID	Category	Requirement	Target/Metric
NFR-DKP-051	Security	Session timeout and warning	Session expires after 30 min inactivity; warning dialog shown 2 min before expiration with option to extend

Chapter 10

On-Premises Infrastructure and DevOps Strategy

10.1 Deployment Objective

MVP (Phase 1): The MVP shall be deployable in university-managed, on-premises infrastructure using self-hosted or locally-available services. Advanced AI features (summarization, plagiarism detection, OCR) shall have self-hosted reference implementations (Hugging Face models, Tesseract, local plagiarism baseline).

Phase 2 (Optional): Cloud API integrations (OpenAI, Google Cloud, Turnitin) may be added as optional enhancements if institutional Memoranda of Understanding (MoU) and network policies permit. No Phase 1 core feature shall require cloud APIs to function.

10.2 On-Premises Reference Architecture (MVP)

Minimum Runtime Components:

- Reverse proxy and TLS termination (Nginx or equivalent)
- Frontend web service
- Backend API service(s)
- Background worker service for async jobs (OCR, summarization, plagiarism)
- PostgreSQL database
- Redis cache and queue broker
- Elasticsearch for search
- Object/file storage for documents and backups

Network Segmentation:

- DMZ/public zone: reverse proxy only
- Application zone: frontend, API, workers
- Data zone: database, search, cache, storage
- Management zone: monitoring, logging, backup/restore tooling

10.3 Infrastructure and Operations Requirements

ID	Category	Requirement	Target/Metric
INF-DKP-001	Compute	Provide isolated environments for dev, staging, and production on-prem	Three separate environments available before pilot
INF-DKP-002	Storage	Separate storage tiers for active content and backups	Backup tier physically/logically isolated from primary storage
INF-DKP-003	Networking	Enforce firewall rules between DMZ, app, data, and management zones	Only required ports open; deny-all default policy
INF-DKP-004	Availability	Use service restart policies and health checks for all core services	Auto-restart enabled; failed service detectable within 1 minute
INF-DKP-005	Backup/DR	Automated daily backup with tested restore procedures	Daily backups; monthly restore drill; $RPO \leq 24h$; $RTO \leq 8h$
INF-DKP-006	Observability	Centralized logs, metrics, and alerting for all services	Alerts for downtime, queue backlog, high error rate, storage thresholds

10.4 Containerization and Environment Parity

ID	Category	Requirement	Target/Metric
DEVOPS-DKP-001	Containerization	All deployable components shall run as versioned container images	Reproducible image build per release tag
DEVOPS-DKP-002	Orchestration	MVP deployment shall support Docker Compose; Phase 2 may adopt Kubernetes	Compose files for dev/staging/prod checked into repository
DEVOPS-DKP-003	Configuration	Environment-specific settings managed via config files and secrets store	No hard-coded secrets in source control
DEVOPS-DKP-004	DB Migration	Schema migrations are automated and version-controlled	Migration step executed in deployment pipeline with roll-back plan

ID	Category	Requirement	Target/Metric
DEVOPS-DKP-005	Security	Container images scanned for vulnerabilities before release	No critical vulnerabilities at release gate

10.5 CI/CD and Release Strategy

ID	Category	Requirement	Target/Metric
REL-DKP-001	CI	Pipeline shall run lint, unit/integration tests, and security scans on every merge request	Pipeline blocks merge on failed quality gates
REL-DKP-002	Artifact Mgmt	Release artifacts shall be tagged and stored in internal registry	Immutable artifacts for each release version
REL-DKP-003	CD	Deployments to staging and production shall be automated and auditable	Deployment logs include who, what version, when, outcome
REL-DKP-004	Rollback	Production deployment must include rollback mechanism	Rollback executable within 15 minutes
REL-DKP-005	Change Mgmt	Database and application changes released using controlled release checklist	No production release without approved checklist

10.6 On-Prem Open Decisions

- Final hardware sizing by expected concurrent users and storage growth
- Selection of internal CI runner platform (GitHub Actions self-hosted, GitLab Runner, Jenkins)
- Selection of monitoring stack (Prometheus/Grafana/ELK or equivalent)
- Backup repository technology and retention period beyond MVP
- Phase 2 decision: Whether to pursue cloud API integrations (OpenAI, Google Cloud, Turnitin) and if so, negotiate institutional MoU before Phase 2 begins
- Phase 2 decision: Network security policy on external API calls from on-prem infrastructure

Chapter 11

Risks and Open Questions

11.1 Risk Register

Risk ID	Description	Prob.	Impact	Mitigation
RISK-DKP-001	Phase 2 cloud API integration may have high costs or rate limits	Low	Medium	MVP uses self-hosted models (no cost); Phase 2 cloud service decisions deferred; MoU negotiation before adoption
RISK-DKP-002	Bangla OCR Character Error Rate (CER) may exceed 15% on benchmark	Medium	Medium	Test Tesseract 4+ and alternatives; benchmark against BanglaOCR dataset; provide manual correction interface
RISK-DKP-003	Plagiarism detection may produce false positives	High	Medium	Set appropriate threshold (25%); manual review for borderline cases
RISK-DKP-004	Integration with existing library system may be complex	Medium	High	Early technical assessment; allocate buffer time; gradual migration
RISK-DKP-006	User adoption may be slow if interface is too complex	Medium	High	User testing throughout development; simplified onboarding; training materials
RISK-DKP-005	File format migrations may lose fidelity	Low	High	Extensive testing; keep originals permanently; manual QA sampling
RISK-DKP-007	Concurrent annotation editing may cause conflicts	Medium	Low	Implement operational transform or CRDT; last-write-wins with notification
RISK-DKP-008	Storage costs may exceed budget with high upload volume	Medium	Medium	Implement quotas per user/department; compression for older files

Risk ID	Description	Prob.	Impact	Mitigation
RISK-DKP-009	Citation network visualization may be slow for large datasets	High	Medium	Limit graph depth; lazy loading; use graph database; client-side rendering
RISK-DKP-010	Copyright issues with uploaded content	Low	High	Clear terms of service; DMCA compliance process; moderator review
RISK-DKP-011	Barcode scanner hardware compatibility issues	Low	Low	Test with common models; support standard HID input; manual entry fallback
RISK-DKP-012	Server infrastructure may be insufficient for peak loads	Medium	High	Load testing before launch; auto-scaling configuration; CDN for static assets
RISK-DKP-013	Data migration from legacy system may have quality issues	High	Medium	Data quality assessment; cleaning scripts; staged migration with rollback
RISK-DKP-014	Security vulnerabilities in third-party libraries	Medium	High	Regular dependency audits; automated vulnerability scanning; update policy
RISK-DKP-015	Feature scope too large to complete in semester timeline	High	High	MVP definition; phased delivery; core features first

11.2 Open Questions

11.2.1 Technical Questions

1. Which AI/ML service for summarization? (OpenAI GPT, Google Cloud NLP, self-hosted BERT/T5)
2. Which Bangla OCR engine provides best accuracy-cost balance?
3. Graph database (Neo4j) for citation network or relational DB?
4. Which plagiarism detection API? (Turnitin, Copyscape, self-hosted)
5. Storage solution: S3-compatible object storage or traditional file system?

11.2.2 Business/Policy Questions

1. Exact loan periods for different item types?
2. Fine calculation: flat rate or progressive? Grace period?

3. Access tier policies: who approves changes?
4. Retention policy for user data after account closure?
5. Who reviews flagged content? Removal process?
6. Institutional repository license: open access requirements? Embargo periods?
7. Maximum file size limits per user role?
8. Acceptable use policy for AI features?
9. Data sharing with external researchers - what are the policies?
10. Intellectual property rights for student projects?

11.2.3 User Experience Questions

1. Should annotations be opt-in or opt-out for sharing?
2. Virtual reading room size limit? (Performance vs. collaboration needs)
3. Email notification frequency: real-time, daily digest, or user-configurable?
4. Should students see each other's projects before approval/publication?
5. Gamification elements: leaderboards public or private? Opt-in?

11.2.4 Integration Questions

1. Single Sign-On (SSO) with institutional identity provider? (LDAP, SAML, OAuth)
2. Integration with existing student information system for auto-enrollment?
3. Export to institutional repository (DSpace, Fedora)? ETD integration?
4. Integration with citation managers: direct API or file export only?
5. Calendar integration for due dates and reading room sessions?

11.2.5 Decisions Needed Before Development

- Confirm institutional server specifications
- Approve technology stack
- Finalize priority ranking (MVP vs. Phase 2)
- Confirm budget for third-party services
- Approve data privacy policy
- Identify pilot user group

Chapter 12

Testability and Traceability

12.1 Requirement Traceability

Sample traceability matrix - full matrix maps all 50 FRs and 50 NFRs:

Requirement ID	Story/Use Case	Planned Test ID
FR-DKP-001	US-DKP-001, UC1	TC-DKP-001, TC-DKP-002
FR-DKP-003	US-DKP-002, UC1	TC-DKP-003, TC-DKP-004
FR-DKP-009	US-DKP-001, UC5	TC-DKP-006, TC-DKP-007
FR-DKP-016	US-DKP-003, UC2	TC-DKP-010, TC-DKP-011
FR-DKP-026	US-DKP-002, UC1	TC-DKP-014
FR-DKP-004	US-DKP-002, US- DKP-006	TC-DKP-005
FR-DKP-005	US-DKP-006	TC-DKP-025
FR-DKP-006	US-DKP-002	TC-DKP-005
FR-DKP-007	US-DKP-002	TC-DKP-014
FR-DKP-010	US-DKP-001, US- DKP-013	TC-DKP-007
FR-DKP-011	US-DKP-006	TC-DKP-008
FR-DKP-012	US-DKP-013	TC-DKP-009
FR-DKP-015	US-DKP-003, US- DKP-011	TC-DKP-010
FR-DKP-017	US-DKP-008	TC-DKP-026
FR-DKP-019	US-DKP-011	TC-DKP-012
FR-DKP-022	US-DKP-003, US- DKP-016	TC-DKP-013
FR-DKP-023	US-DKP-016	TC-DKP-013, TC-DKP-023
FR-DKP-024	US-DKP-006	TC-DKP-025
FR-DKP-025	US-DKP-008	TC-DKP-027
FR-DKP-027	US-DKP-010	TC-DKP-010
FR-DKP-028	US-DKP-015	TC-DKP-014
FR-DKP-029	US-DKP-007, UC1	TC-DKP-015
FR-DKP-031	US-DKP-020, UC5	TC-DKP-015
FR-DKP-032	US-DKP-005	TC-DKP-028
FR-DKP-033	US-DKP-014	TC-DKP-016
FR-DKP-035	US-DKP-009	TC-DKP-017

Requirement ID	Story/Use Case	Planned Test ID
FR-DKP-036	US-DKP-017	TC-DKP-017
FR-DKP-037	US-DKP-004, UC3	TC-DKP-018, TC-DKP-019
FR-DKP-040	US-DKP-018, UC3	TC-DKP-020
FR-DKP-041	US-DKP-012	TC-DKP-028
FR-DKP-046	US-DKP-019, UC4	TC-DKP-021
FR-DKP-048	UC4	TC-DKP-021
NFR-DKP-001	All use cases	TC-DKP-022
NFR-DKP-002	US-DKP-001	TC-DKP-006, TC-DKP-022
NFR-DKP-011	US-DKP-001	TC-DKP-023
NFR-DKP-013	All use cases	TC-DKP-023
NFR-DKP-019	All use cases	TC-DKP-022
NFR-DKP-028	All user stories	TC-DKP-024
NFR-DKP-029	All user stories	TC-DKP-024
NFR-DKP-032	All user stories	TC-DKP-034
NFR-DKP-040	US-DKP-001, US-DKP-013	TC-DKP-007, TC-DKP-009

12.2 Initial Manual Test Plan

Minimum 15 test cases required - 25 provided. All test cases detailed below:

Test ID	Scenario	Steps & Expected Result
TC-DKP-001	User registration with institutional email	Steps: 1) Navigate to registration 2) Enter name, valid institutional email 3) Enter password (8+ chars, complex) 4) Submit 5) Check email verification. Expected: Account created with 'member' role; verification email received; link activates account
TC-DKP-003	Upload PDF document	Steps: 1) Navigate to Upload 2) Drag PDF file ; 500MB 3) Wait for upload. Expected: Progress bar shows 100%; file preview displayed; metadata extraction begins
TC-DKP-010	Checkout book with barcode	Steps: 1) Scan member ID 2) Verify member details 3) Scan book barcode 4) Review due date 5) Confirm. Expected: Checkout record created; book status "Checked Out"; email confirmation sent; due date is 14 days from today
TC-DKP-014	Submit research paper	Steps: 1) Navigate to Submit Paper 2) Upload PDF 3) Review auto-filled metadata 4) Wait for plagiarism check 5) Review AI summary 6) Submit. Expected: Paper uploaded; plagiarism score ; 25%; AI summary generated; paper in "Under Review" state

Test ID	Scenario	Steps & Expected Result
TC-DKP-018	Create annotation on PDF	Steps: 1) Open PDF 2) Select text 3) Click Annotate 4) Enter comment 5) Set visibility Private 6) Save. Expected: Text highlighted; annotation saved; visible only to user; persists on reload
TC-DKP-002	Login with valid credentials	Steps: 1) Navigate to login 2) Enter correct email 3) Enter correct password 4) Click Login. Expected: User redirected to dashboard; session created; user profile visible in header
TC-DKP-004	Upload unsupported file format	Steps: 1) Navigate to Upload 2) Attempt to upload .exe file. Expected: Upload rejected; error message displays allowed formats
TC-DKP-005	Edit document metadata	Steps: 1) Open document in metadata editor 2) Edit Title field 3) Add keywords 4) Change language to Bangla 5) Save. Expected: Metadata saved; auto-save indicator shows timestamp; changes persist
TC-DKP-006	Bilingual search - English	Steps: 1) Enter "machine learning" in search 2) Press Enter. Expected: Results display documents containing the term; ranked by relevance; filters work
TC-DKP-007	Bilingual search - Bangla	Steps: 1) Switch to Bangla 2) Enter Bangla search term 3) Press Search. Expected: Results show matching Bangla documents; properly displays Unicode
TC-DKP-008	Bangla OCR on scanned document with benchmark	Steps: 1) Load BanglaOCR benchmark dataset (50 test images) 2) Upload each test Bangla PDF 3) Run OCR extraction 4) Calculate Character Error Rate (CER) per image. Expected: CER $\leq$ 15% on benchmark set; manual correction UI available; performance log generated
TC-DKP-009	Voice search in English	Steps: 1) Click microphone icon 2) Allow microphone 3) Speak "digital library" 4) Stop. Expected: Speech converts to text; search executes; results displayed
TC-DKP-011	Renew borrowed book	Steps: 1) Navigate to My Account ; Current Loans 2) Find book 3) Click Renew 4) Confirm. Expected: Due date extended by 14 days; renewal count incremented; email sent
TC-DKP-012	Place hold on checked-out book	Steps: 1) Search for checked-out book 2) Open details 3) Click Place Hold 4) Confirm. Expected: Hold request created; queue position shown; email confirmation
TC-DKP-013	Return book with QR scan	Steps: 1) Navigate to Returns 2) Scan book QR code 3) Confirm return 4) Select condition 5) Complete. Expected: Loan updated with return date; book status "Available"; fine calculated if overdue

Test ID	Scenario	Steps & Expected Result
TC-DKP-015	AI summarization with ROUGE-L benchmark	Steps: 1) Load CNN/DailyMail or academic benchmark (20 papers with reference summaries) 2) Submit each paper to summarization engine 3) Calculate ROUGE-L score against reference summary 4) Aggregate results. Expected: Mean ROUGE-L ≥ 0.35 ; individual scores logged; user edit/reject UI functional
TC-DKP-016	Plagiarism detection on submission	Steps: 1) Upload document in submission flow 2) Wait for check 3) View report. Expected: Plagiarism score shown; warning if $\geq 25\%$; similar sections highlighted
TC-DKP-017	Generate APA citation	Steps: 1) Open paper details 2) Click Cite 3) Select APA 4) Copy to clipboard. Expected: Citation formatted in APA style; includes all required fields; copies successfully
TC-DKP-019	Collaborative annotation sharing	Steps: 1) Open annotated document 2) Change visibility to Study Group 3) Share link 4) Group member opens. Expected: Annotations visible to group; group member can reply; real-time sync
TC-DKP-020	Create virtual reading room	Steps: 1) Navigate to Reading Rooms 2) Click Create 3) Enter name and topic 4) Invite members 5) Add document 6) Create. Expected: Room created; invitations sent; shared document accessible; chat active
TC-DKP-021	Digital preservation health check	Steps: 1) Navigate to Preservation Dashboard 2) Click Run Health Check 3) Wait for scan 4) Review results. Expected: Checksum verification runs; obsolescence warnings displayed; report generated
TC-DKP-022	Performance: Page load time	Steps: 1) Clear cache 2) Navigate to home page 3) Use DevTools to measure 4) Repeat 3 times. Expected: Average load time ≤ 2 seconds; DOM content loaded ≤ 1 second
TC-DKP-023	Security: Password complexity	Steps: 1) Try password "1234" 2) Try "password" 3) Try "Pass123!" 4) Complete registration. Expected: First two rejected with specific message; third accepted; account created
TC-DKP-024	Accessibility: Keyboard navigation	Steps: 1) Load home page 2) Press Tab repeatedly 3) Navigate menus with Enter/Arrow 4) Submit search. Expected: All elements reachable; focus indicators visible; no keyboard traps
TC-DKP-025	Bulk upload with CSV	Steps: 1) Navigate to Bulk Upload 2) Download template 3) Upload filled CSV with 50 items 4) Review preview 5) Import. Expected: CSV validated; 50 items previewed; import succeeds; items searchable

Test ID	Scenario	Steps & Expected Result
TC-DKP-026	Overdue notification email	Steps: 1) Set item due date to tomorrow 2) Wait for system job 3) Check email next day. Expected: Email sent 1 day before due date; contains item details and due date
TC-DKP-027	Generate librarian report	Steps: 1) Navigate to Librarian Reports 2) Select "Overdue Items" 3) Set date range 4) Generate 5) Export to Excel. Expected: Report displays overdue items; member contact info shown; Excel export works
TC-DKP-028	Citation network visualization	Steps: 1) Open paper with 20 citations 2) View citation network 3) Filter by year 4) Click node. Expected: Interactive graph loads; shows citing/cited papers; filters work; node clickable
TC-DKP-029	Export annotations to PDF	Steps: 1) Open annotated document 2) Select Export Annotations 3) Choose PDF 4) Download. Expected: PDF generated with all comments; includes context; properly formatted
TC-DKP-030	Session timeout after inactivity	Steps: 1) Login 2) Wait 30 minutes without activity 3) Try to access protected page. Expected: Session expires; user redirected to login; warning shown 2 min before expiration
TC-DKP-031	Responsive design on mobile	Steps: 1) Open site on mobile (320px width) 2) Navigate menu 3) Perform search 4) View document. Expected: Layout responsive; hamburger menu works; touch targets 44x44px; all features accessible
TC-DKP-032	Screen reader announces content	Steps: 1) Enable screen reader (NVDA) 2) Load page 3) Navigate with arrows. Expected: Main content announced; semantic HTML used; ARIA labels present; form labels associated
TC-DKP-033	User data deletion with audit log anonymization	Steps: 1) Admin initiates user deletion request 2) System marks user profile for deletion 3) System anonymizes audit log entries (hash user ID, remove identifiable fields) 4) Verify user personal data removed, audit logs retained with hashed references. Expected: User account deleted; documents re-assigned or archived; audit logs present but identifiers anonymized; deletion completed within 30 days

Chapter 13

Submission Checklist

Problem statement and scope are clear

Minimum 12 user stories with acceptance criteria (20 provided)

Functional requirements table is complete (50 requirements)

Data model and data dictionary are attached

Minimum 5 wireframes or flow screens included (8 wireframes)

NFR section is completed (50 non-functional requirements)

Risks and open questions are documented (15 risks, 25+ open questions)

Initial manual test plan (15+ cases) is included (32 test cases)

Chapter 14

Appendices

14.1 Glossary

Access Tier Permission level (public, member, staff, restricted) determining who can view content

Annotation User-created note or highlight on a document

Bangla NLP Natural Language Processing for Bangla language (stemming, synonyms)

Citation Network Graph showing relationships between papers through citations

Digital Preservation Long-term storage and accessibility of digital content

DOI Digital Object Identifier - unique persistent identifier for documents

Lifecycle State Status of document (draft, review, published, archived)

MARC Machine-Readable Cataloging - library metadata standard

OCR Optical Character Recognition - extracting text from images

Repository Organized collection of documents and research outputs

14.2 Technology Stack Recommendations

14.2.1 Frontend

- Framework: React.js
- UI Library: Material-UI or Ant Design
- PDF Viewer: PDF.js or React-PDF
- Annotation: Annotator.js or Hypothesis
- Charts: D3.js or Chart.js

14.2.2 Backend

- Language: Node.js (Express)
- API: RESTful + GraphQL
- Authentication: JWT + OAuth 2.0
- Background Jobs: Bull

14.2.3 Database

- Primary: PostgreSQL
- Search: Elasticsearch (Bangla support)
- Cache: Redis
- Graph: Neo4j (citation network)

14.2.4 AI/ML Services

MVP (Phase 1) - Self-Hosted Implementations

- Summarization: Hugging Face BART or T5 (self-hosted or on-device)
- NLP: spaCy with Bangla models; gensim for topic modeling
- OCR: Tesseract 4+ with Bangla language data
- Plagiarism Detection: Local sequence matching baseline (mandatory) + optional integration hooks

Phase 2 (Optional) - Cloud API Integrations

If institutional MoU permits and network access is available:

- Summarization Enhancement: OpenAI GPT-4 (premium option)
- NLP Enhancement: Google Cloud Natural Language (premium option)
- OCR Enhancement: Google Cloud Vision with Bangla support (premium option)
- Plagiarism Enhancement: Turnitin or Copyleaks API (premium option)

Note: Phase 2 cloud integrations are optional and require explicit institutional approval and service agreements.

14.3 Conclusion

The Digital Knowledge Platform represents a comprehensive solution to fragmented digital resource management in academic institutions. By integrating digital archives, research repositories, library management, and student project showcases, we address critical needs for efficient knowledge discovery, collaboration, and preservation.

14.3.1 Unique Competitive Advantages

1. **AI-Powered Intelligence:** Research assistance and automated summarization
2. **Bilingual Excellence:** Deep Bangla language support (OCR, NLP, voice)
3. **Collaborative Knowledge Building:** Real-time annotation and virtual reading rooms

4. **Academic Integrity Focus:** Built-in plagiarism detection and citation tools
5. **Digital Preservation:** Proactive format health monitoring
6. **Comprehensive Analytics:** Data-driven insights and research impact metrics

14.3.2 Success Metrics

- User adoption: 70% of target users active within 3 months
- Efficiency gains: 50% reduction in time to find resources
- Research visibility: 30% increase in downloads and citations
- Library operations: 40% faster circulation processing
- User satisfaction: 80% satisfaction score
- System performance: Meeting all NFR targets

This SRS provides a solid foundation for building a world-class digital knowledge platform that will serve our institution for years to come.

END OF DOCUMENT